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COURSE: CHE 312: Transport Phenomena II

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Quiz #1

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Closed Notes and Book

Duration: 20 minutes

The rear window of an automobile is defogged by passing warm air at 40°C over its inner surface, and the associated convection coefficient is $30\text{ W}/(\text{m}^2\cdot\text{K})$. Under conditions for which the outside ambient air temperature is -20°C and the associated convection coefficient is $65\text{ W}/(\text{m}^2\cdot\text{K})$, what are the inner and outer surface temperatures of the window? The window glass is 4 mm thick and the thermal conductivity of glass is $1.4\text{ W}/(\text{m}\cdot\text{K})$.

40°C inner, $30\text{ W}/(\text{m}^2\cdot\text{K})$, -20°C , $65\text{ W}/(\text{m}^2\cdot\text{K})$
4 mm, $1.4\text{ W}/(\text{m}\cdot\text{K})$

$$\frac{40^\circ\text{C} - (-20^\circ\text{C})}{\frac{30\text{ W}}{\text{m}^2\cdot\text{K}} + \frac{1.4\text{ W}}{0.004\text{ m}\cdot\text{K}} + \frac{65\text{ W}}{\text{m}^2\cdot\text{K}}} = \frac{(-(-20)) \cdot 1.4 (40 - 20)}{4T_{\text{ico}} - T_{\text{oo}}}$$

$40 - (-20) = 60$

$$\frac{40^\circ\text{C} - 20^\circ\text{C}}{1.4\text{ W}/(\text{m}\cdot\text{K}) \cdot 0.004} = \frac{293\text{ K}}{0.046\text{ W/K}}$$

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